RandomEig

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1 Random Matrix Eigenvalues

Question: let M be an n-dimensional, square, complex-entry matrix. Each of the entries is a (uniformly) random number $a_{ij} \in [-1, 1]$. Study its properties

1.1 Diagoanlizability

Theorem: these matrices will always be diagonalizable, we need not worry about that

Proof: let $P(M)$ be the characteristic polynomial of the matrix. Due to the fundamental theorem of algebra, this polynomial has exactly n complex roots. Therefore, the matrix is diagonalizable (as a complex matrix)$\blacksquare$

1.2 Constrains

Before analyzing the ins and outs of the problem, it may be helpful to analyze its constrains. Since the matrices are random in the initial basis but we are interested in their eigenvalues (i.e. in their form in their eigenbasis), we shall base this study on base-independent quantities, such as the trace and the determinant.

1.2.1 1. Trace constrains

Proposition: $Tr[M]$ is a uniform distribution between $[-1, 1]$, just as its entries

Proof: $Tr[M] = \sum_i a_{ii}$, so it is a sum of uniform distributions, and hence a uniform distribution itself, between the same limits

Proposition: the sum of all eigenvalues will be a real quantity and its maximum (in abs.) will be n: $\boxed{\text{abs}(\lambda_1 + ... + \lambda_n) \leq n}$

Proof: the highest trace the original matrix can have is achieved by having all 1s in the diagonal (the lowest, with all -1s). This adds up to n. Since the trace is basis-independent, the trace of the diagonal matrix (i.e. the sum of all eigenvalues) will be real and equal to $\pm n$ at best$\blacksquare$

Proposition: if the distribution were between $[0, 1]$, then the maximum (real) eigenvalue we would get is $\lambda_{max} = n$. However, this would entail the diagonal of M to be composed entirely of 1s, and therefore the probability of getting a maximum eigenvalue falls with the dimension of the matrix, as for every dimension we need one more entry in the diagonal to be a 1: $P(\lambda = \lambda_{max}) = [P(a_{ii} = 1)]^n$

This is less obvious when we allow eigenvalues to adopt negative values, as they can counter the positive ones. Therefore, if we have $\lambda_1 = 500+n$ and $\lambda_2 = -500$ in a 2x2 matrix, then we would still satisfy the trace condition. However, experiment suggests the eigenvalues are constrained inside a quasi-circumference which is much more bounded than simply a circumference of radius n (e.g. in $n = 50$ they are contained inside a $R = 4$ circumference)

Proposition: even in the $[-1, 1]$ case, there still exists the possibility that $\lambda = n$. It appears to be wildly suppressed

Proof: let us consider a 2x2 matrix with entries in $[-1, 1]$, such that its rows are not linearly independent, for instance: $M = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Upon diagonalizing it, we get: $D = \text{diag}(2, 0)$, so indeed $\exists \lambda = n$. The generalization to higher dimensions is trivial, so long as all the spectrum is 0 but one of the eigenvalues $\blacksquare$

1.3 2. Determinant constrains

Def.: let us define the determinant of a matrix as a complex function $\det : \mathfrak{M}_n \mapsto \mathbb{C}$

Proposition: the product of the eigenvalues of M will always be real and less than unity in absolute value: $\boxed{\text{abs}(\lambda_1 \ldots \lambda_n) \leq n}$

Proof: the maximum determinant a 2x2 matrix can have is $\det \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = 2$. The generalization to higher dimensions is immediate, essentially you want to have all 1s with a sign equal to the parity of the permutation. Since the determinant is basis-invariant, the determinant of the diagonal basis (i.e. the product of the eigenvalues) will be the same, so even if the eigenvalues are complex, their product will be real and capped (in abs) at $n \blacksquare$

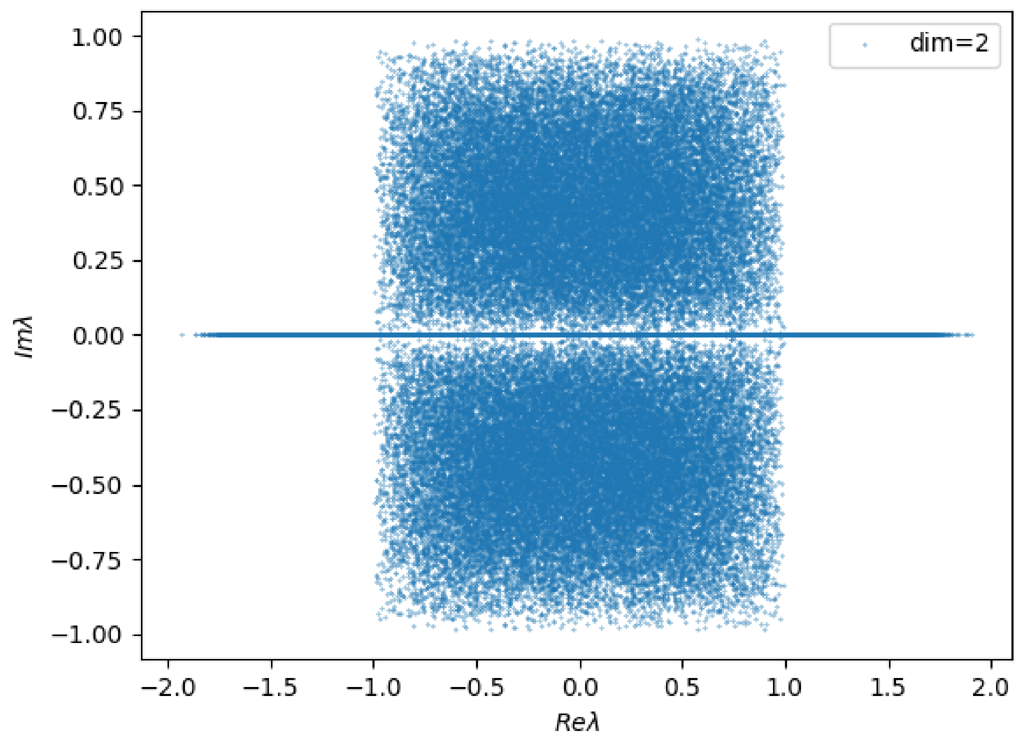
2 Q1: why are the complex eigenvalues confined inside a region?

Approach:

1. First constrain the real eigenvalues
2. Then study the region of the complex plane where the probability of finding an eigenvalue is higher than some threshold value. Perhaps it would be more interesting to develop a position density function for the eigenvalues in the complex plane

Observation: the shape of the complex eigenvalue cloud depends on the dimension. It would, at first, appear to depend on the number of samples taken, but for a large enough sample you see there are some soft limits for the region. I say they are soft because there always seem to be some eigenvalues which are outside the hard region but these are a minority

Examples: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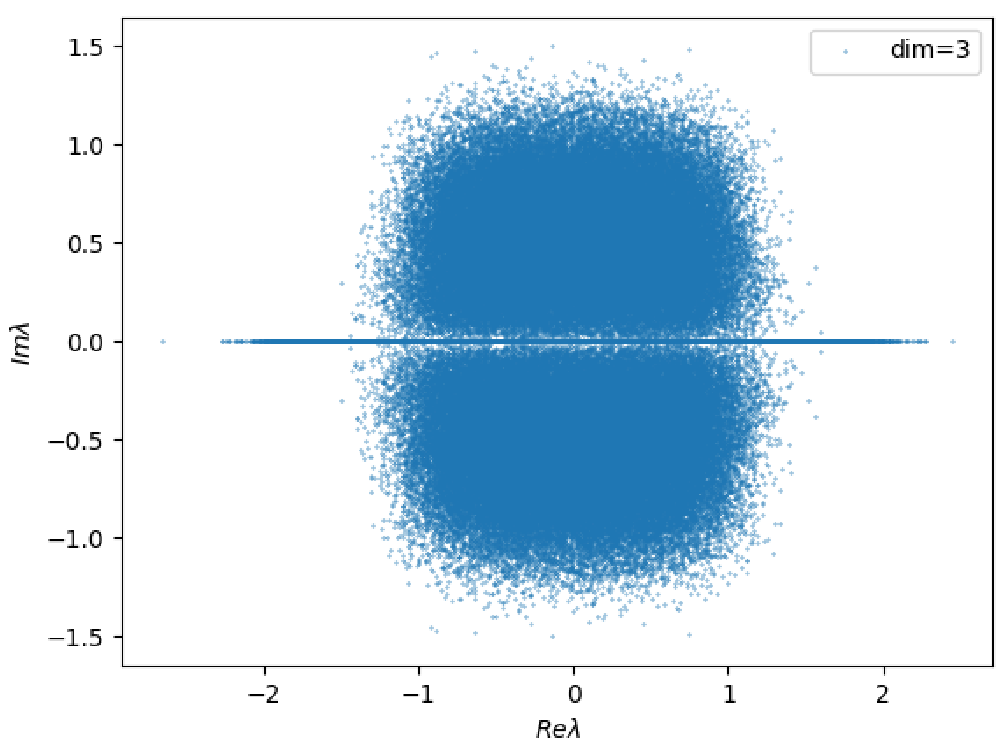
0.0
0.5
1.0
1.5
2.0
Reλ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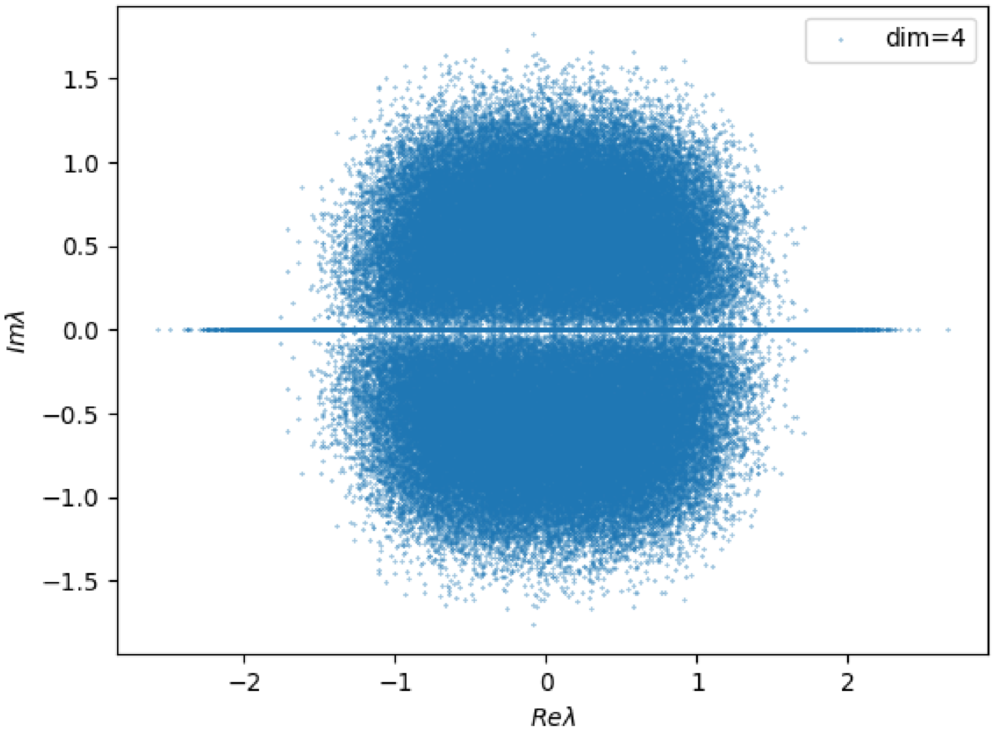
2
Reλ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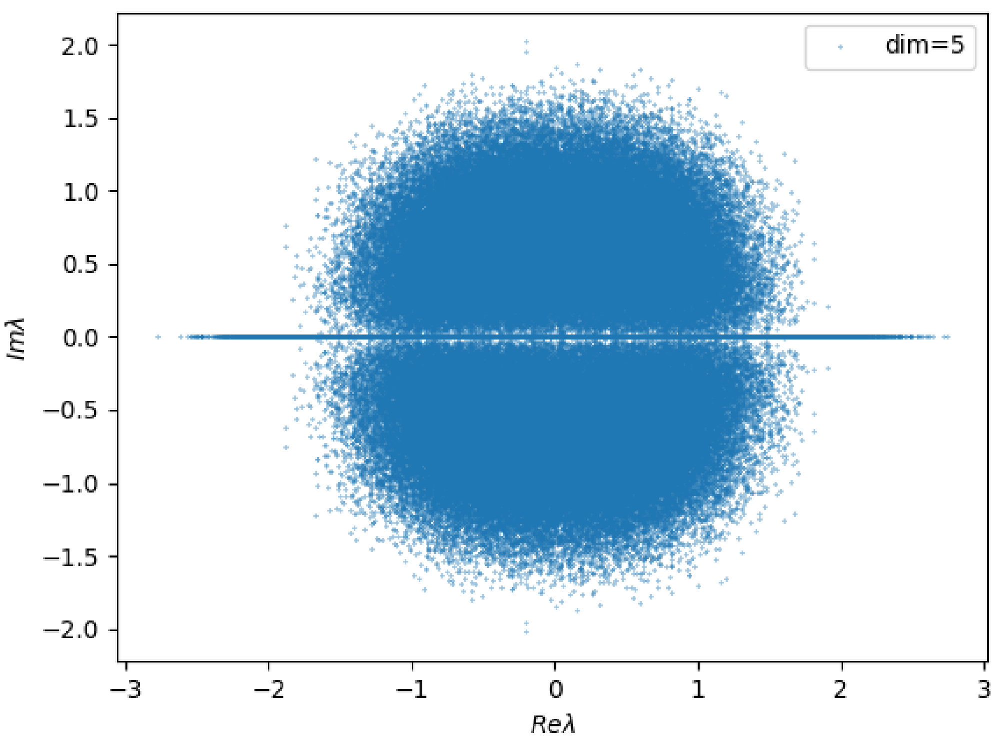
2
$Re\lambda$
$Im\lambda$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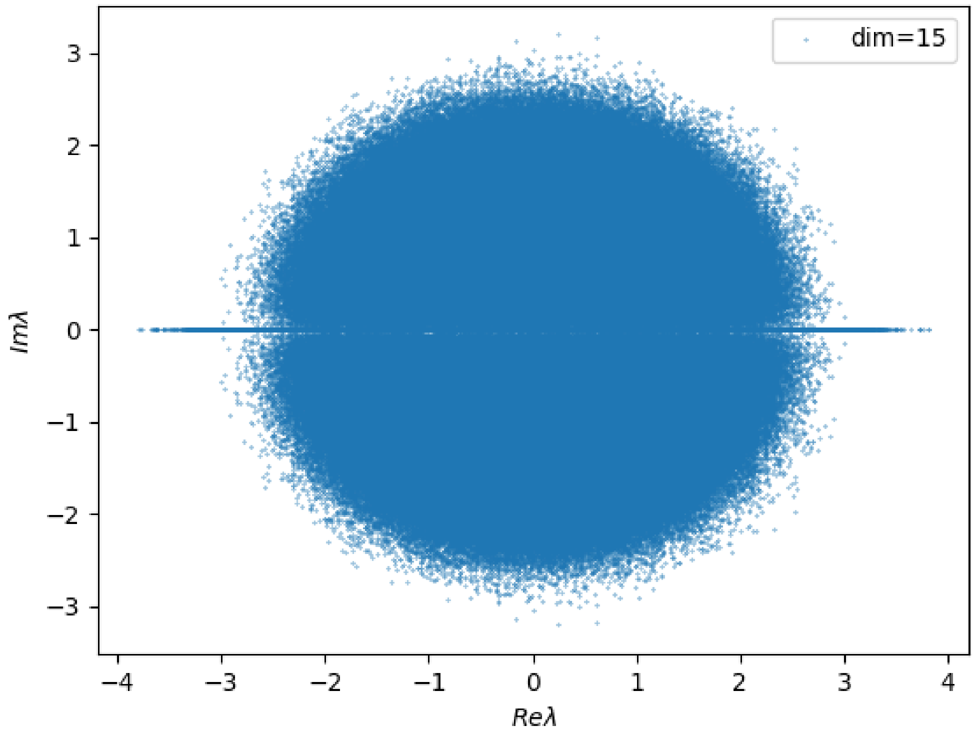
1
2
3
$Re\lambda$
$Im\lambda$



Observation: as we increase the dimension of the matrices, the hard region evolves from a rectangle to a circumference

2.1 Probability density

Def.: let $f : \mathbb{C} \to \mathbb{R}$ be a probability density function, such that it maps the point density for the $N_{repetitions} \to \infty$ limit in a simulation

Normalization: $\int_{\mathbb{C}} f(\lambda) d\lambda = 1$. I could normalize the point density to give me all the points used in the simulation (i.e. all the eigenvalues calculated), but this would be divergent in the thermodynamic limit, as $N = n \times N_rep$. I could also normalize it to n, but this choice is arbitrary and for that purpose I may as well normalize it to 1 so it is a true probability density. The interpretation of this normalization would then be "from all the eigenvalues calculated, give me the fraction contained in all of the complex plane."

Advantage of this normalization condition: even if the number of eigenvalues calculated is infinite, the fraction of eigenvalues contained in a region, calculated as $\chi_\Omega \equiv \int_\Omega f(\lambda) d\lambda,\ \Omega \subset \mathbb{C}$ is less than 1, because $f(\lambda)$ is a probability density

Miracle: knowing $f(\lambda)$ in all of the complex plane would solve all our questions: it would predict the supression of small imaginary part-ed eigenvalues (i.e. repulsion), it would also naturally constrain the region of the complex plane on which we find the bulk of eigenvalues, and it would predict that, although having a maximal eigenvalue ($\lambda = n$) is possible, in practice it is wildly suppressed by dimensionality for $n \geq 3$, so the point density in these regions is much smaller (although strictly nonzero, we would still get an infinite number of points here!) than in the much likelier bulk

Procedure: how can we determine $f(\lambda)$? –> either from direct calculation or from pheno, if the former fails

Q2: why is there a repulsion of the eigenvalues with small imaginary part?